

BALKASHINO, Kazakhstan

WMO: 289780

Lat: 52.5283N

Lon: 68.7428E

Elev: 398

StdP: 96.63

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.4	-30.1	-36.4	0.1	-33.3	-33.0	0.2	-29.8	11.8	-9.0	10.7	-9.0	1.3	230	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.3	17.3	28.5	16.9	26.6	16.5	19.4	26.3	18.6	25.5	17.8	24.4	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.8	21.8	16.2	12.1	21.0	15.3	11.4	20.3	57.1	26.3	54.4	25.5	51.8	24.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.8	8.4	-37.0	33.2	2.9	1.7	-39.1	34.4	-40.8	35.4	-42.4	36.3	-44.5	37.5		
(5)			-37.1	21.0	2.9	0.8	-39.2	21.6	-40.8	22.0	-42.4	22.5	-44.5	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.7	-17.0	-15.2	-7.7	3.6	11.8	16.7	17.8	16.5	10.6	3.4	-6.8	-13.8		
	DBStd	13.91	7.17	6.69	6.19	6.18	4.84	4.18	3.41	4.16	4.74	5.12	7.30	7.21		
	HDD10.0	3846	837	707	549	204	38	4	0	3	49	213	504	739		
	HDD18.3	6158	1095	940	807	442	208	78	50	85	237	464	754	997		
	CDD10.0	830	0	0	0	12	93	204	243	205	66	7	0	0		
	CDD18.3	100	0	0	0	0	4	28	35	29	3	0	0	0		
	CDH23.3	1406	0	0	0	4	111	408	415	387	81	1	0	0		
	CDH26.7	391	0	0	0	0	21	117	111	119	22	0	0	0		
	WSAvg	3.5	3.0	3.7	3.9	3.8	3.9	3.4	3.1	3.0	3.1	3.4	3.7	3.7		
(15) Precipitation	PrecAvg	404	23	19	22	24	39	42	75	45	25	32	30	27		
	PrecMax	632	52	34	52	55	89	94	185	129	63	67	69	63		
	PrecMin	249	2	1	2	2	4	14	24	8	4	6	10	7		
	PrecStd	98	12	8	14	14	23	22	41	34	14	16	15	15		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	0.1	6.9	23.2	29.2	32.0	32.4	32.4	29.8	21.3	8.2	0.2		
		MCWB	-0.6	-0.7	2.7	12.3	14.7	17.8	18.3	17.7	15.2	10.5	5.6	-0.7		
	2%	DB	-2.3	-1.7	3.6	19.7	26.3	29.7	29.6	29.8	25.6	17.6	5.3	-0.9		
		MCWB	-2.7	-2.3	1.4	10.2	14.1	16.8	17.9	17.4	13.5	9.1	3.8	-1.3		
	5%	DB	-4.3	-3.4	1.7	16.7	24.0	27.9	27.4	27.6	22.1	14.6	3.2	-2.5		
		MCWB	-4.8	-4.0	0.4	9.1	13.0	16.9	17.6	17.1	12.4	7.8	1.9	-3.1		
	10%	DB	-6.2	-5.2	0.4	13.5	21.8	25.6	25.6	25.2	19.3	11.6	1.4	-4.1		
		MCWB	-6.7	-5.8	-0.5	7.7	12.4	16.0	17.4	16.3	11.2	6.6	0.5	-4.6		
	0.4%	WB	-0.6	-0.6	3.7	13.3	16.9	19.8	20.6	20.0	16.8	11.7	6.4	-0.2		
		MCDB	-0.4	0.0	6.0	20.8	23.7	27.6	26.5	27.3	26.8	18.6	7.5	0.0		
	2%	WB	-2.6	-2.2	1.7	11.2	15.1	18.7	19.6	18.9	14.6	10.0	3.8	-1.3		
		MCDB	-2.3	-1.7	3.1	17.6	23.2	26.4	26.1	26.1	22.1	15.5	5.2	-1.0		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-4.8	-4.0	0.6	9.6	14.2	17.8	18.9	18.0	13.2	8.4	2.1	-3.0		
		MCDB	-4.4	-3.5	1.7	15.6	22.1	25.4	25.5	25.4	20.6	13.6	3.0	-2.6		
	10%	WB	-6.8	-5.9	-0.4	8.0	13.1	16.8	18.1	17.0	11.9	7.0	0.5	-4.6		
		MCDB	-6.3	-5.2	0.5	13.4	20.3	23.8	23.9	23.7	17.8	11.1	1.2	-4.1		
		MDBR	8.4	9.1	9.0	10.6	13.4	12.9	11.7	12.9	12.4	9.5	6.9	7.6		
		MCDBR	7.0	8.0	7.8	16.1	17.1	16.5	15.4	16.9	17.1	15.6	7.4	7.2		
(36) Mean Daily Temperature Range	5% DB	MCWBR	7.1	7.6	6.2	8.5	7.6	6.8	6.2	6.7	7.6	8.7	6.1	7.0		
		MCDBR	7.2	7.3	6.6	14.3	14.6	14.2	12.8	14.3	15.4	13.4	6.8	7.1		
	5% WB	MCWBR	7.3	7.2	5.6	8.4	7.5	6.5	6.1	6.6	7.4	8.1	5.9	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.281	0.310	0.364	0.400	0.397	0.399	0.418	0.410	0.358	0.339	0.313	0.280		
	taud		2.079	2.044	1.963	2.135	2.241	2.309	2.244	2.250	2.351	2.296	2.140	2.061		
	Ebn at Noon		679	774	800	828	853	854	828	812	818	745	631	597		
	Edh at Noon		75	106	142	137	129	122	128	121	97	83	71	64		
(44) All-Sky Solar Radiation	RadAvg		1.25	2.38	3.79	4.71	5.77	6.09	5.64	4.90	3.57	1.91	0.91	0.84		
	RadStd		0.09	0.12	0.42	0.46	0.51	0.52	0.42	0.38	0.26	0.27	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page